Ahmad Hossein Yazdani

CONTACT INFORMATION

Department of Computer Science
Faculty of Engineering
Virginia Polytechnic Institute and State University
E-mail: ahmadyazdani@vt.edu
Website: *https://ayazdani1997.github.io/*

AREAS OF INTEREST

Design and Analysis of Algorithms, Artificial Intelligence, Machine Learning, Deep Learning, Storage Systems, File Systems, Kernel Programming, Enterprise Application Design, Computer Networks, Neuroscience, Adversarial Machine Learning, Natural Language Processing (NLP), Network Security, Back-end Development, Distributed computing, High performance computing, Cloud computing, Distributed Systems, High Performance Computing

EDUCATION

1. Virginia Polytechnic Institute and State University
   PhD. In Computer Science, August 2020-present

2. University of Tehran, Tehran, Iran
   Bachelor of Science in Computer Software Engineering, September 2015-July 2020

3. *Diploma at Allameh Helli High School,* June 2014

RESEARCH EXPERIENCE

Research Assistant at Distributed System and Storage Lab at Virginia Tech. university
Supervised by Dr. Ali Butt
More information on https://dssl.cs.vt.edu/index.html
August 2020-present

RESEARCH PROJECTS

1. *Stock price prediction using deep learning, finished in July 2020*
   Under supervision of Professor Nasser Yazdani (As my undergraduate final project).
   The code is available through GitHub

2. *Metis project as a research project that I have contribution to. It aims to do I/O caching for deep learning frameworks. It's still in progress.*

NOTABLE PROJECTS

1. TOORLA Programming Language Compiler Development over JVM

2. ATALK Actor-based Programming Language Compiler Development over MIPS

3. Design and Implementation of House Rental Enterprise Application as Internet Engineering University Course Project by ReactJS and both Java Enterprise Application and NodeJS

4.. TCP Protocol Development under UDP as Computer Network University Course Project

5. Adding extra Synchronization mechanism to Linux kernel as Operating System University Course Project

6. MIPS Architecture Implementation on Cyclone II FPGA

7. Meeting Scheduler Enterprise Application as Software Engineering University Course Project

8. Design and Implementation of online Shopping Enterprise Application with Waterfall Software Development Methodology (Responsible for the back-end part by means of NodeJS)

9. Image classification using Spiking Neural Networks as my final Advanced Machine Learning course at Virginia Tech.

10. Development of a centralized cache on top of docker

NOTABLE PASSED COURSES

- **Undergraduate Courses**

1. Engineering Mathematics
2. Artificial Intelligence
3. Engineering Probability and Statistics
4. Computer Networks
5. Computer Workshop
6. Object Oriented Systems Design
7. Human Computer Interaction
8. Compiler Design and Implementation
9. Database Design
10. Computer Aided Digital System Design
11. Internet Engineering
12. Machines and Language Theory
13. Software Engineering
14. Parallel Programming
15. Computer Architecture Laboratory
16. Introduction to Software Testing
17. Formal Methods in Software Engineering
18. Object Oriented Systems Design
19. Human Computer Interaction
20. Computer Architecture
21. Microprocessors
22. Systems Analysis and Design
23. Calculus 1
24. Calculus 2
25. B.SC Project

- **Graduate Courses**

1. Advanced Machine Learning
2. Graduate Operating System

TEACHING ASSISTANTSHIP

Teaching Assistantship of Programming Languages and Compilers
University of Tehran
Fall 2018 - winter 2019
Presented by Dr. Fatemeh Ghassemi & Dr. Hossein Hojjat
As Homework Organizer and Course Project Grading TA

Teaching Assistantship of Programming Languages and Compilers
University of Tehran
Winter 2018 - spring 2019,
Presented by Dr. Hossein Hojjat
As Course Project Organizer

Teaching Assistantship of Formal Methods in Software Engineering
University of Tehran
Winter 2019 - spring 2020,
Presented by Dr. Fathieh Faghih
As both homework and project organizer

Teaching Assistantship of Artificial Intelligence
University of Tehran
Winter 2020 – summer 2020,
Presented by Dr. Hakimeh Fadaei
As both homework and project organizer
(Organized the project associated with Neural Networks and Deep Learning)

Teaching Assistantship of CS1114 (Introduction to Software Design)
Virginia Tech.
Fall 2020 – winter 2020,
Presented by Dr. Dwight Barnette
As Lab TA and Programming Assignment TA

ATTENDED CONTESTS

Attending Sharif University of Technology *contest hold in summer 2017*
Attending codeforces contest *hold in 2017*

INTERNSHIP EXPIERNECE

Internship at Router lab at University of Tehran *in summer 2017*
Supervised by Dr. Nasser Yazdani
More information on https://digt.ir

Internship at Router lab at University of Tehran *in summer 2018*
Supervised by Dr. Nasser Yazdani

LANGUAGE SKILLS

Persian
English

CERTIFICATES

TOEFL, GRE

FAVORITE PROGRAMMING LANGUAGES And

Programming Languages: C, C++, Java, Python, Go, HTML, CSS, JavaScript
HDL: Verilog, VHDL
Applications: Microsoft Office
Database: MySQL, SQLite, MS SQL

FRAMEWORKS

Machine Learning libraries and frameworks: sklearn, Pytorch, Tensorflow
Enterprise Application frameworks: Django, ReactJS, NodeJS
Containerization frameworks: Docker
Computer Network frameworks: GNS3
Document preparation: Latex

HOBBIES

Watching movies, Reading books, Quick Text Typing, Football, Chess

REFERENCES

Dr Ali Butt:
butta@vt.edu

Dr. Hoda Eldardiry:
hdardiry@vt.edu

Dr. Ahmad Khonsari:
aa_khonsari@ut.ac.ir

Dr. Hossein Hojjat:
hojjat@ut.ac.ir

Dr. Fatemeh Ghassemi:
fghassemi@ut.ac.ir

Dr. Faghih Fathieh:
f.faghih@ut.ac.ir

Dr. Nasser Yazdani:
yazdani@ut.ac.ir

LINKS

1. https://www.linkedin.com/in/ahmad-hossein-yazdani-6560b3138/ *(LinkedIn profile)*

2. *https://github.com/Ayazdani1997 (GitHub profile)*

3. *https://gitlab.com/A.Yazdani (GitLab profile)*